

Multiple Choice (10 marks)

1. In a _____ attack, the extra data that holds some specific instructions in the memory for actions is projected by a cyber-criminal or penetration tester to crack the system.
 - (a) Phishing
 - (b) MiTM
 - (c) Buffer-overflow
 - (d) Clickjacking
2. Which of the following is not an example of presentation layer issues?
 - (a) Poor handling of unexpected input can lead to the execution of arbitrary instructions
 - (b) Unintentional or ill-directed use of superficially supplied input
 - (c) Cryptographic flaws in the system may get exploited to evade privacy
 - (d) Weak or non-existent authentication mechanisms
3. The stack is memory for storing
 - (a) Local variables
 - (b) Program code
 - (c) Dynamically linked libraries
 - (d) Global variables
4. When a wireless user authenticates to any AP, both of them go in the course of four-step authentication progression which is called _____
 - (a) AP-handshaking
 - (b) 4-way handshake
 - (c) 4-way connection
 - (d) wireless handshaking
5. What are the types of scanning?
 - (a) Port, network, and services
 - (b) Network, vulnerability, and port
 - (c) Passive, active, and interactive
 - (d) Server, client, and network

True / False (10 marks)

Answer True or False

6. True or False: A packet filter firewall filters data at the application or transport layer of the OSI model.
7. True or False: The digest created by a hash function is known as a Message Authentication Cipher.
8. True or False: A Base Signal Station is used by mobile operators to offer signal coverage and is equivalent to a Base Transceiver Station.
9. True or False: Message confidentiality ensures that only the intended sender and receiver can access the content of a communication.
10. True or False: To verify a digital signature, the sender's public key is required to authenticate the signature's validity.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the difference between the squared sum of first n natural numbers and the sum of squared first n natural numbers.
12. Write a function to check whether the given string is ending with only alphanumeric characters or not using regex.

End of Paper